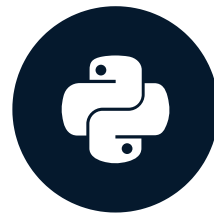


Types of joins

PANDAS JOINS FOR SPREADSHEET USERS

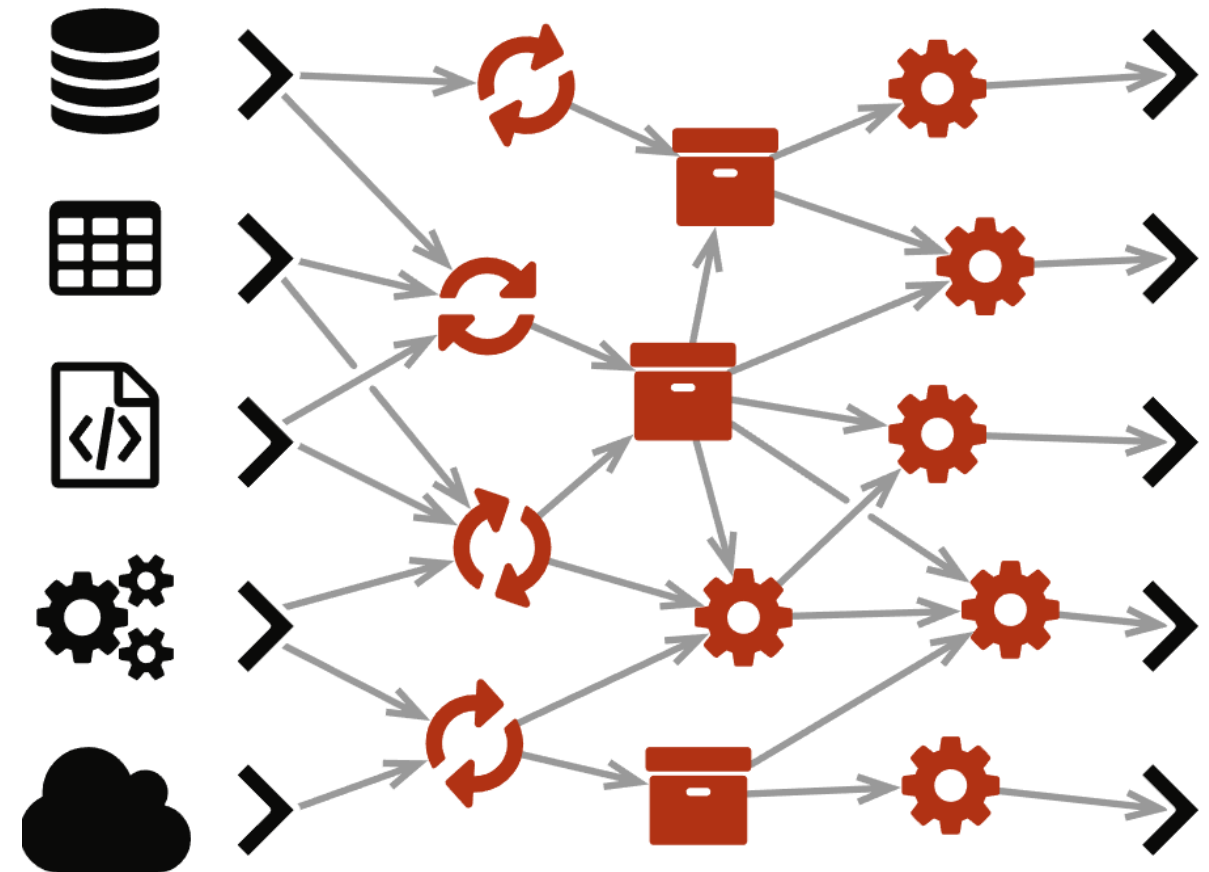


John Miller

Principal Data Scientist

Three types of joins

- Types
 - One-to-one
 - One-to-many
 - Many-to-many
- Join type indicates relationship of tables
- Like lookup values in VLOOKUP



One-to-one

	A	B	C
1	GameKey	Season	SeasonType
2	1	2016	Pre
3	2	2016	Pre
4	3	2016	Pre
5	4	2016	Pre
6	5	2016	Pre
7	6	2016	Pre
8	7	2016	Pre
9	8	2016	Pre

	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of F
3	2	Grass	Los Angeles Memoria
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field
6	5	Grass	Soldier Field
7	6	Grass	Heinz Field
8	7	Natural grass	Levis Stadium
9	8	A-Turf Titan	Ralph Wilson Stadium



	A	B	C	D	E
1	GameKey	Season	SeasonType	Turf	Stadium
2	1	2016	Pre	Turf	Tom Benson Hall of F
3	2	2016	Pre	Grass	Los Angeles Memoria
4	3	2016	Pre	Natural Grass	M&T Bank Stadium
5	4	2016	Pre	DD GrassMaster	Lambeau Field
6	5	2016	Pre	Grass	Soldier Field
7	6	2016	Pre	Grass	Heinz Field
8	7	2016	Pre	Natural grass	Levis Stadium
9	8	2016	Pre	A-Turf Titan	Ralph Wilson Stadium

- Datasets at the same level
- Lookup column same for both tables
- Rows match one-to-one

One-to-many

	A	B	C
1	GameKey	Season	SeasonType
2	2	2016	Pre
3	3	2016	Pre
4	4	2016	Pre
5	5	2016	Pre
6	6	2016	Pre
7	7	2016	Pre
8	8	2016	Pre
9	9	2016	Pre

	A	B	C
1	GameKey	PlayId	GameClock
2	2	191	12:30
3	2	1132	12:08
4	3	1676	1:51
5	3	2643	6:10
6	3	3949	1:59
7	4	291	9:32
8	4	386	8:37
9	4	927	1:53



	A	B	C	D	E
1	GameKey	Season	SeasonType	PlayId	GameClock
2	2	2016	Pre	191	12:30
3	2	2016	Pre	1132	12:08
4	3	2016	Pre	1676	1:51
5	3	2016	Pre	2643	6:10
6	3	2016	Pre	3949	1:59
7	4	2016	Pre	291	9:32
8	4	2016	Pre	386	8:37
9	4	2016	Pre	927	1:53

- Rows are unique at different levels
- Join based on the higher-level column
- Result based on lower-level column
- Similar to VLOOKUP based on lower-level data

Many-to-many

	A	B
1	PlayerId	GameKey
2	32689	9
3	32689	350
4	32689	581
5	33024	618
6	33024	597
7	33699	252
8	33699	554
9	33699	560
10	33699	560

	A	B	C
1	PlayerId	Number	Position
2	32689	85	TE
3	32689	49w	TE
4	32689	49o	TE
5	33024	98	DE
6	33024	93	DE
7	33024	67	DE
8	33024	63	DE
9	33699	03d	RB
10	33699	30	RB



	A	B	C	D
1	PlayerId	GameKey	Number	Position
2	32689	9	85	TE
3	32689	9	49w	TE
4	32689	9	49o	TE
5	32689	350	85	TE
6	32689	350	49w	TE
7	32689	350	49o	TE
8	32689	581	85	TE
9	32689	581	49w	TE
10	32689	581	49o	TE

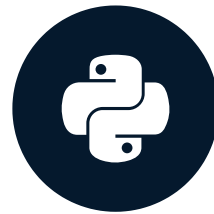
- Values are repeated
- Results in all possible combinations
- Can be hard to interpret
- Best supplemented with a "bridge" table

Let's practice!

PANDAS JOINS FOR SPREADSHEET USERS

A closer look at one-to-one joins

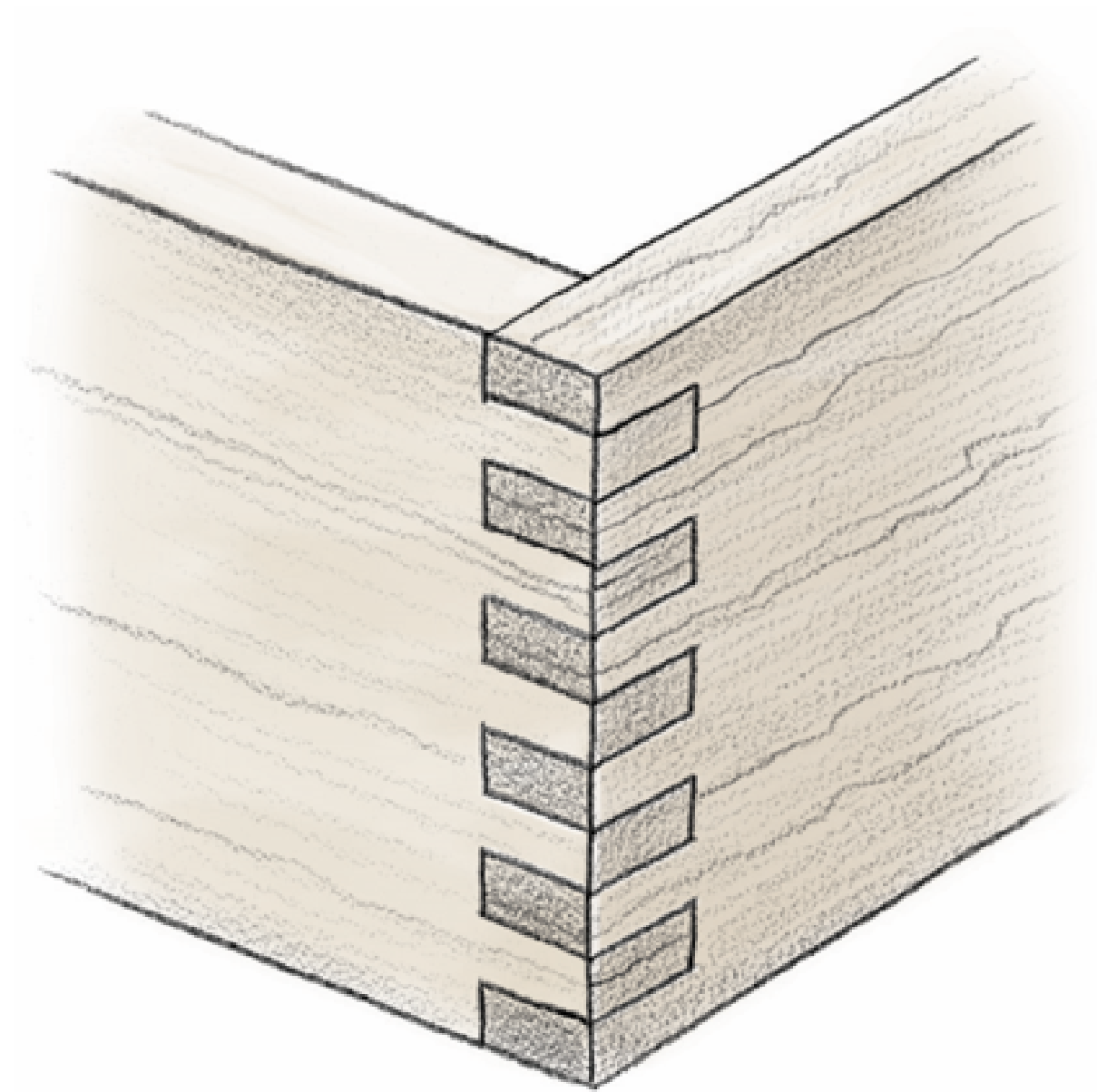
PANDAS JOINS FOR SPREADSHEET USERS



John Miller
Principal Data Scientist

Basics of pandas.merge()

- Used with complementary data
- `pd.merge(left, right)`
- Joins on one or more columns
- Similar to VLOOKUP



Left merges

	A	B
1	GameKey	Season
2	2	2016
3	4	2016
4	6	2016
5	8	2016

	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of Fame
3	2	Grass	Los Angeles Memorial Col
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field
6	5	Grass	Soldier Field
7	6	Grass	Heinz Field
8	7	Natural grass	Levis Stadium
9	8	A-Turf Titan	Ralph Wilson Stadium

↓

	A	B	C	D
1	GameKey	Season	Turf	Stadium
2	2	2016	Grass	Los Angeles Memorial
3	4	2016	DD GrassMaster	Lambeau Field
4	6	2016	Grass	Heinz Field
5	8	2016	A-Turf Titan	Ralph Wilson Stadium

```
pd.merge(df_left, df_right,  
         on='GameKey', how='left')
```

- List data frame names for "df_left" and "df_right" placeholders
- Resulting frame matches rows in left frame

Right merges

	A	B
1	GameKey	Season
2	2	2016
3	4	2016
4	6	2016
5	8	2016

	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of Fame
3	2	Grass	Los Angeles Memorial Col
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field
6	5	Grass	Soldier Field
7	6	Grass	Heinz Field

	A	B	C	D
1	GameKey	Season	Turf	Stadium
2	1		Turf	Tom Benson Hall of Fame Stadi
3	2	2016	Grass	Los Angeles Memorial Coliseur
4	3		Natural Grass	M&T Bank Stadium
5	4	2016	DD GrassMaster	Lambeau Field
6	5		Grass	Soldier Field
7	6	2016	Grass	Heinz Field

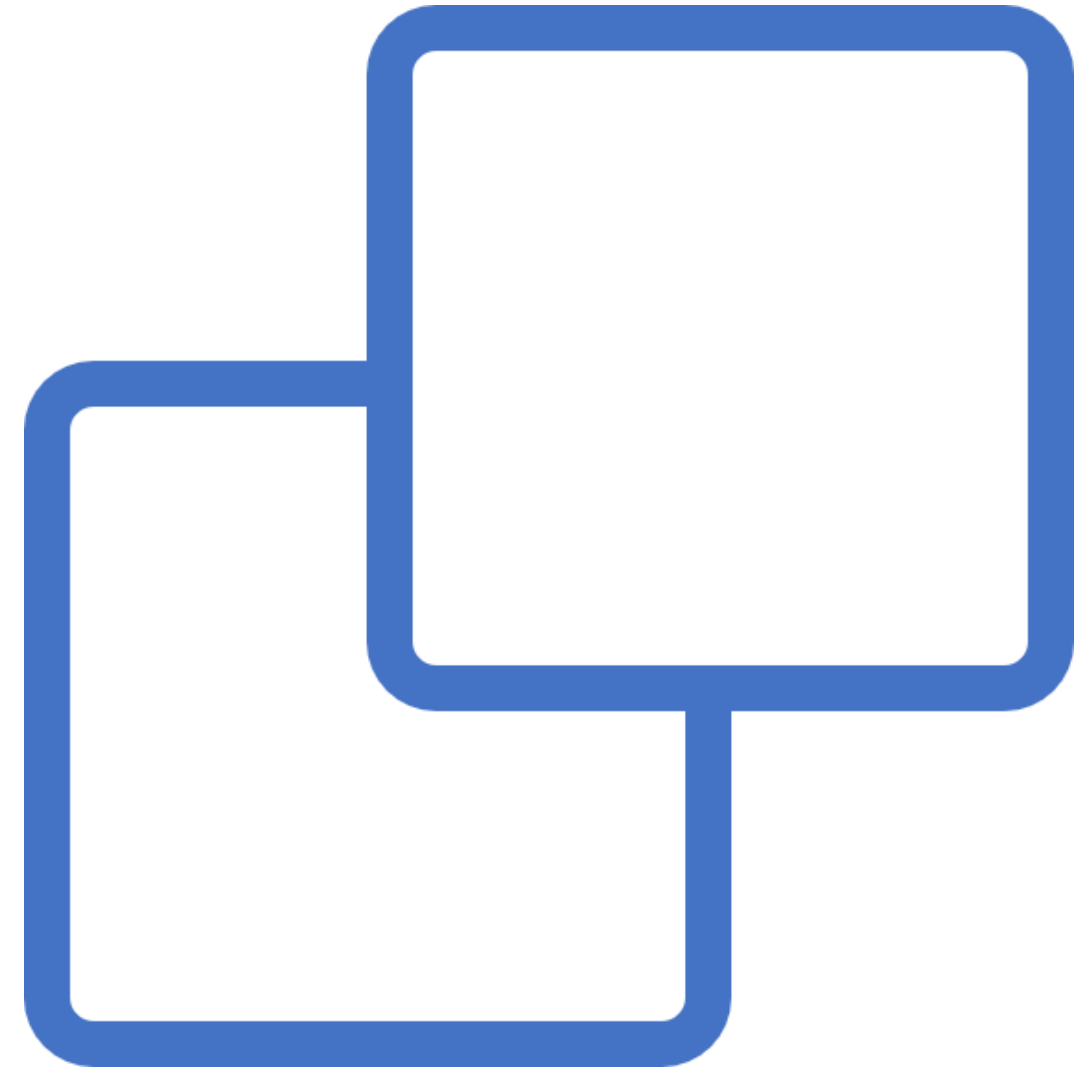
```
pd.merge(df_left, df_right,  
         on='GameKey', how='right')
```

- Same concept as left merge
- Resulting frame rows match right frame

A framework for joins

After viewing and understanding the data:

- Determine the relationship
- Check for unique values in key column
- Write merge statement and execute



Joining on two keys

	A	B	C	D
1	GameId	PlayId	FieldPos	YardLine
2	2017090700	100	NE	35
3	2017090700	101	NE	43
4	2017090700	102		
5	2018090000	100		
6	2018090000	101		

	A	B	C	D
1	GameId	PlayId	Quarter	GameClock
2	2017090700	100	1	0.14:14:
3	2017090700	101	1	0.13:52:
4	2017090700	102	1	0.13:02:
5	2018090000	100	1	0.12:12:
6	2018090000	101	1	0.12:08:

- Determine the relationship
- Check for unique values in key column
- Write merge statement and execute

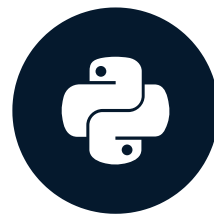
```
pd.merge(df_left, df_right,  
         on=['GameKey', 'PlayId'])
```

Let's practice!

PANDAS JOINS FOR SPREADSHEET USERS

Combining common data with inner joins

PANDAS JOINS FOR SPREADSHEET USERS



John Miller
Principal Data Scientist

Object-oriented expressions

pandas-level function:

```
pd.merge(df_left, df_right, on=None)
```

Shorthand, pandas dataframe method:

```
df_left.merge(df_right, on=None)
```

A basic inner join

	A	B
1	GameKey	Season
2		2
3	4	2016
4	6	2016
5	8	2016

	A	B	C
1	GameKey	Turf	Stadium
2		3 Natural Grass	M&T Bank Stadium
3	4	DD GrassMaster	Lambeau Field
4	5	Grass	Soldier Field
5	6	Grass	Heinz Field

↓

	A	B	C	D
1	GameKey	Season	Turf	Stadium
2	4	2016	DD GrassMaster	Lambeau Field
3	6	2016	Grass	Heinz Field

- Similar to VLOOKUP followed by removing NA's

```
df1.merge(df2, on='GameKey', how='inner')
```

- Returns only rows present in both tables

Joining on different names

	A	B
1	GameKey	Season
2	2	2016
3	4	2016
4	6	2016
5	8	2016

	A	B	C
1	game-key	Turf	Stadium
2	3	Natural Grass	M&T Bank Stadium
3	4	DD GrassMaster	Lambeau Field
4	5	Grass	Soldier Field
5	6	Grass	Heinz Field

↓

	A	B	C	D
1	GameKey	Season	Turf	Stadium
2	4	2016	DD GrassMaster	Lambeau Field
3	6	2016	Grass	Heinz Field

- Key columns may have different names
- No need to rename columns

```
df1.merge(df2, left_on='GameKey', right_on='game-key',  
          how='inner')
```

Let's practice!

PANDAS JOINS FOR SPREADSHEET USERS